3/3/26

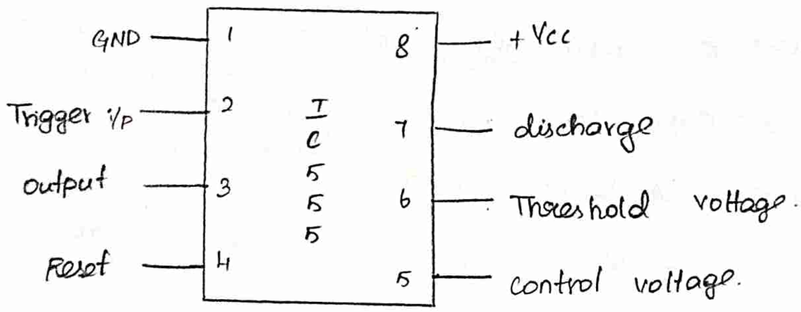
UNIT-IV.

TIMER AND PLL :

1) IC 555 timer.
2). Voltage controlled oscillator (VCO)
3). Phase locked loop (PLL)

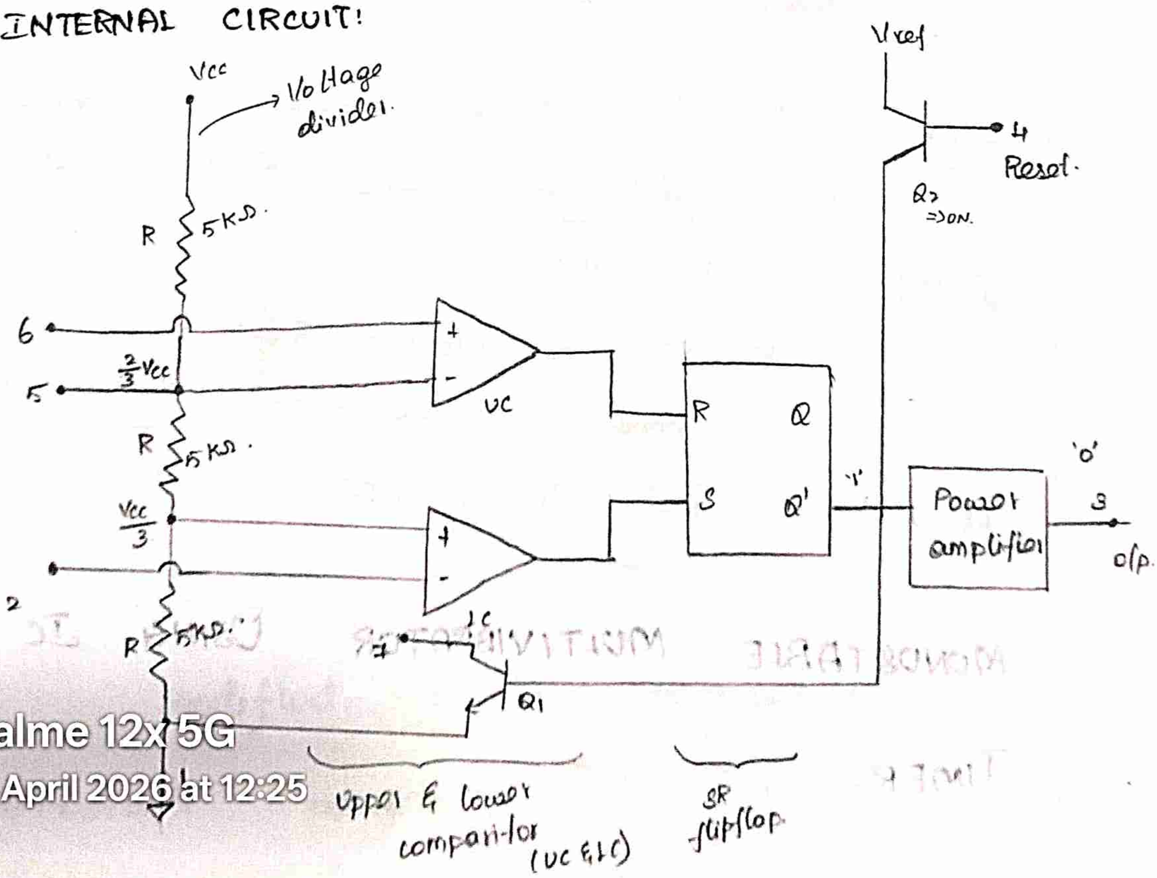
IC 555 Timer:

PIN DIAGRAM.



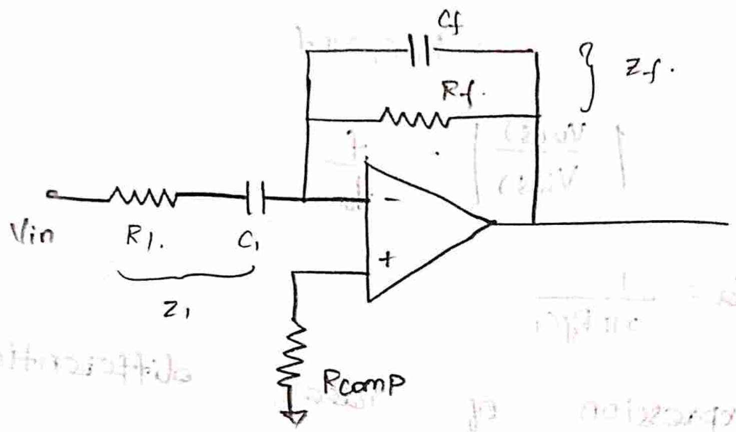
Supply voltage is 4.8 to 18 V.

INTERNAL CIRCUIT!



⇒ to improve the stability we add one capacitor in feedback.

Practical differentiator:



$$Z_f = R_f \parallel \frac{1}{sC_f}, \quad Z_1 = R_1 + \frac{1}{sC_1}$$

$$\frac{V_o}{V_{in}} = -\frac{Z_f}{Z_1}$$

$$Z_1 = \frac{sR_1C+1}{sC_1}.$$

$$\frac{1}{Z_f} = \frac{1}{R_f} + \frac{1}{1/sC_f}.$$

$$= \frac{1}{R_f} + sC_f.$$

$$= \frac{1+sR_fC_f}{R_f}.$$

$$Z_f. = \frac{R_f}{1+sR_fC_f}.$$

$$\frac{V_o}{V_{in}} = -\frac{\frac{R_f}{1+sR_fC_f}}{\frac{1+sR_1C_1}{sC_1}}$$

Gain, $\frac{V_o(s)}{V_i(s)} = -R_f C_1 (j\omega)$.

$$\left|\frac{V_o(s)}{V_i(s)}\right| = R_f C_1 \omega$$

$$= R_f C_1 2\pi f.$$

$$\left|\frac{V_o(s)}{V_i(s)}\right| = \frac{f}{f_a}.$$

where, $f_a = \frac{1}{2\pi R_f C_1}$

* Gain expression of ideal differentiation.

<u>Limitation of ideal differentiator:</u>

⇒ when (i/p) signal frequency ↑se what (about)

(is) the → i/p impedance.

→ gain.

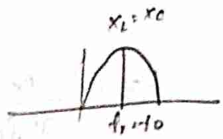
* i/p frequency ↑se - gain is more, it leads <u>unstability</u>.

* for high i/p frequency - i/p impedance is very low bcoz of this the circuit will be affected by <u>high frequency noise</u>.

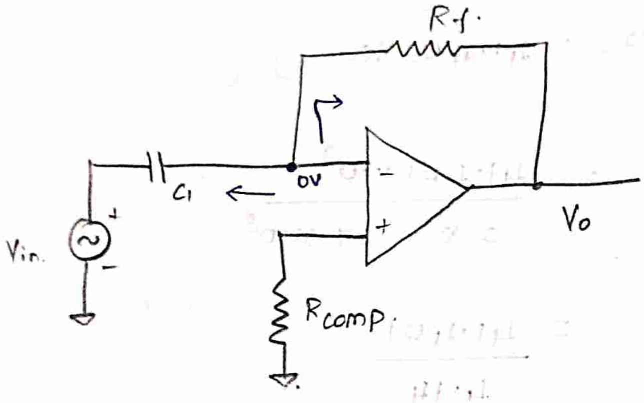
⇒ To overcome this we add Resistor, so that the minimum impedance is 'R' when high frequency. It is not affected by High freq noise.

Resonance - at which the gain is max.

$(X_L = X_C)$



Differentiator :



$\Rightarrow$ It differentiates the applied i/p.

Apply Nodal equation ;

$$\frac{0 - V_{in}}{(1/sC_1)} + \frac{0 - V_o}{R_f} = 0 .$$

$$\frac{V_o(s)}{R_f} = -sC_1 V_{in}(s)$$

$$V_o(s) = -sR_f C_1 V_{in}(s)$$

$$\therefore \frac{V_o(s)}{V_i(s)} = -R_f C_1 s$$

Inverse laplace of V_o ..

$$L^{-1}[V_o(s)] \Rightarrow \boxed{V_o(t) = -R_f C_1 \frac{dV_{in}(t)}{dt}}$$

* Expression of ideal differentiator.

$$\boxed{R_2 = 63.111\,\Omega.}$$

4). $R_3 = \dfrac{Q}{\pi f_0 C} = \dfrac{13}{3.14\times 1\times 10^{3}\times 10^{-7}}$

$= 4.1401\times 10^{4}$

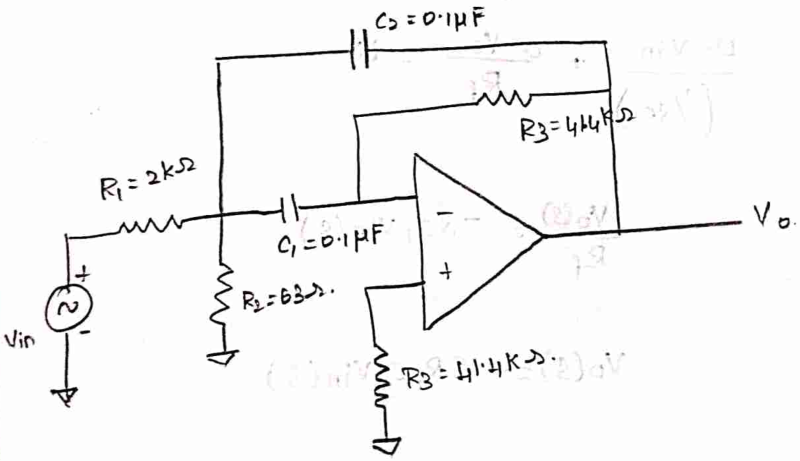
$$\boxed{R_3 = 41.401\,k\Omega.}$$

5). $A_F = \dfrac{R_3}{2R_1} = \dfrac{41.401\times 10^{3}}{2\times 2.07\times 10^{3}}$

$= \dfrac{41.401}{4.14}$

$= 10.000$

$$\boxed{A_F = 10}$$



Find R_2' from f_0 - 1kHz to 10kHz.

$R_2' = R_2\left(\dfrac{f_0}{f_0'}\right)$

$= 63.111 \times \dfrac{1\times 10^{3}}{10\times 10^{3}}$

$$\boxed{R_2' = 6.311\,\Omega.}$$

3). $R_2 = \dfrac{Q}{2\pi f_0 C\left[2Q^2 - A_F\right]}$

4). $R_3 = \dfrac{Q}{\pi f_0 C}$

5). $A_F = \dfrac{R_3}{2R_1}$

6). $R_2' = R_2\left[f_0/f_0'\right]$

$f_0 \rightarrow$ initial value of centre frequ

change f_0

$f_0' \rightarrow$ final value.

change $R_2 \rightarrow R_2'$

also $A_F < 2Q^2$.

(4). Design a NBPF with centre frequency of 1KHz, $Q = 13$ and $A_F = 10$ also assume $C = 0.1\mu F$.

Sol:- Given, $f_0 = 1KHz$, $Q = 13$, $A_F = 10$.

1). $C_1 = C_2 = 0.1\ \mu F$.

2). $R_1 = \dfrac{Q}{2\pi f_0 C A_F} = \dfrac{13}{2 \times 3.14 \times 1 \times \cancel{10^3} \times 10^{\cancel{-7}\ -6\ -3} \times 10}$

$= \dfrac{13}{6.28}$

$\boxed{R_1 = 2.07\ K\Omega}$

3). $R_2 = \dfrac{Q}{2\pi f_0 C\left[2Q^2 - A_F\right]} = \dfrac{13}{6.28 \times \cancel{10^3} \times 10^{\cancel{-7}\ 4}\left[2 \times 169 - 10\right]}$

$= \dfrac{13}{6.28 \cdot (328) \times 10^{-4}}$.

$= \dfrac{13}{2059.84} \times 10^4$.

$= 6.3111 \times 10^{\cancel{-4}\ 4} \times 10^{3}$

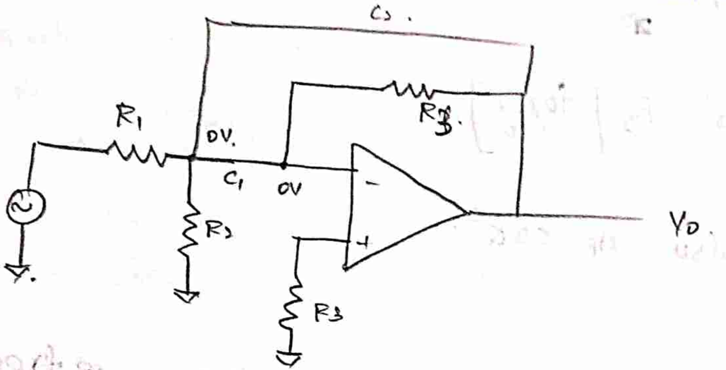
Case (ii):

Consider $f_{in} > f_H$

X_{C1} & $X_{C2} = 0$ [it act as closed switch/circuit].

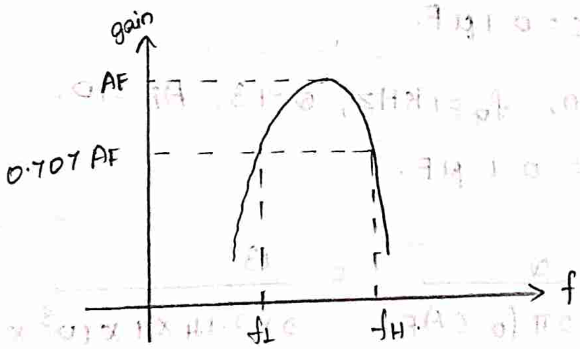
$\boxed{V_0 = 0}$

=> It exhibits the behaviour of LPF.



Case (iii):

consider, $f_L < f < f_H$.



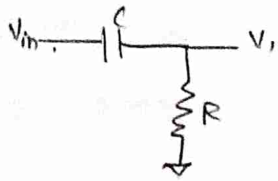
Advantage:

* By changing R_2 we can transient the center frequency of NBPF from one value to another value.

Steps: Design parameters for NBPF:

1). Assume $C_1 = C_2 = C$.

$$R_1 = \frac{Q}{2\pi f_o C A_F} = \frac{Q}{2\pi f_c \cdot C \cdot A_F}.$$



$$V_1 = \frac{V_{in}}{R + 1/SC} \cdot R$$

$$V_1 = V_{in} \cdot \left(\frac{RSC}{1+SRC} \right) \rightarrow (1)$$

$$V_o = \left(1 + \frac{R_f}{R_1} \right) V_1$$

$$= A_F \cdot \frac{V_{in}(SRC)}{1+SRC}$$

$$\frac{V_o}{V_{in}} = A_F \cdot \frac{SRC}{1+SRC} \qquad \because S = j\omega, \ \omega = 2\pi f$$

$$\frac{V_o}{V_{in}} = A_F \cdot \left(\frac{j2\pi fRC}{1+j2\pi fRC} \right)$$

$$\boxed{\frac{V_o}{V_{in}} = \frac{A_F \cdot j(f/f_L)}{1+j(f/f_L)}}$$

here, $f_L = \frac{1}{2\pi RC}$

$f_L \rightarrow$ cut-off frequency of HPF.

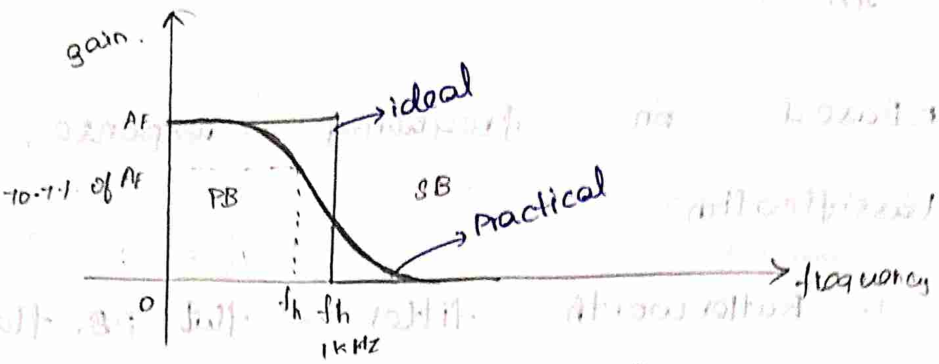
<u>case (i):</u> $f \ll f_L$

$$\left| \frac{V_o(s)}{V_i(s)} \right| = \frac{A_c (f/f_L)}{\sqrt{1 + (f/f_L)^2}}$$

[$\because$ It's gain is very less]

$$\left| \frac{V_o(s)}{V_i(s)} \right| < A_F \rightarrow \text{reduced.}$$

$\therefore$ It attenuates all the signal frequency, $f \ll f_L$



For ideal characteristics of LPF,

PB = 100% gain. [we get exact transition]

SB = No gain.

For Practical,

*. We can't get exact transition from high to low.

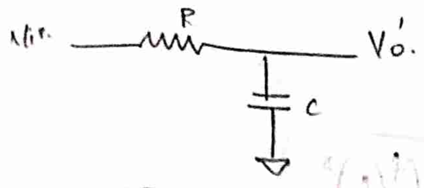
*. It is monotonically decreasing (Gain) as frequency ↑so.

f_h → frequency at which the separation (or) transition from PB to SB. [ideal]. [cutoff / corner freq].

f_h → It is frequency at which the gain reduced by 3 dB.

The gain which is reduced to 70.7% of it's maximum gain, that frequency cut-off frequency. [Practical].

First order butterworth active LPF :



Passive LPF

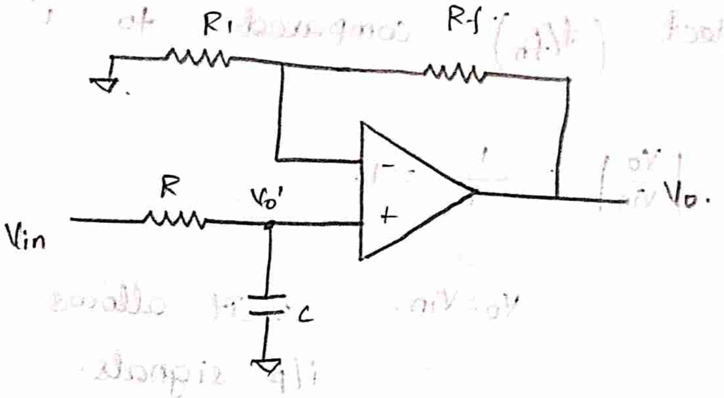
$$V_o' = \frac{V_{in}}{R + 1/sC} \cdot \frac{1}{sC}.$$

$$V_o' = \frac{V_{in}}{1+sRC}.$$

$$A = \frac{V_o'}{V_{in}} = \frac{1}{1+j\omega RC}$$

=> max gain is, $A = 1$, when $(\omega = 0)$

=> when $\omega = \infty$, $A = 0$.



$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_o' \longrightarrow ①.$$

Advantage:

=> In order to improve the gain of passive LPF we added active component. that advantage of active component is applied to. active filter.

=> By choosing proper R_f & R_1 we can improve the gain.

$$V_o = A_F \cdot \frac{V_{in}}{1+sRC}.$$

$$= \frac{A_F}{1+j\omega RC} \cdot V_{in}.$$

$$= \frac{A_F}{1+j2\pi fRC} \cdot V_{in}.$$

Advantage: 2m

1. No loading effect in active filter.
2. small size compare to passive filter.
3. low cost bcoz [inductance is not involved].

$$\frac{V_o}{V_{in}} = \frac{A_F}{1+j(f/f_h)} \longrightarrow ⑤.$$

where, $f_h = \frac{1}{2\pi RC}$.

$$\left|\frac{V_o}{V_{in}}\right| = \frac{1}{\sqrt{1+\left(f/f_h\right)^2}}$$

case (i): $f << f_h$.

Neglect (f/f_h) compared to '1'

$$\left|\frac{V_o}{V_{in}}\right| = \frac{1}{1} = 1.$$

$V_o = V_{in}$. ∴ It allows all the i/p signals.

case (ii): $f >> f_h$.

neglect 1, $1+\left(f/f_h\right)^2 \approx \left(f/f_h\right)^2$

$$\left|\frac{V_o}{V_{in}}\right| = \frac{1}{\sqrt{\left(f/f_h\right)^2}} = \frac{1}{f/f_h}.$$

$$\left|\frac{V_o}{V_{in}}\right| = \frac{f_h}{f}$$

∴ signal is get attenuated /deselected.

It supress the all i/p signal.

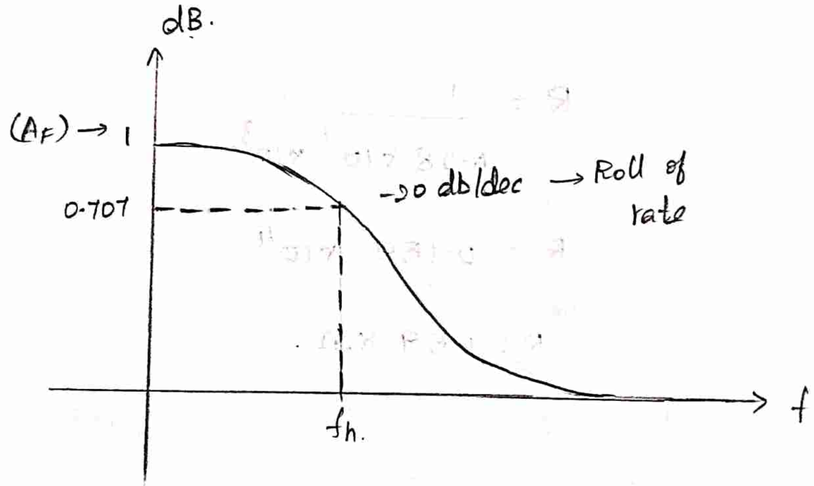
case (iii) :

$$f = f_h .$$

$$\left| \frac{V_o}{V_{in}} \right| = \frac{1}{\sqrt{1+(1)^2}} = \frac{1}{\sqrt{2}} .$$

$$\left| \frac{V_o}{V_{in}} \right| = 0.707$$

The gain of circuit is reduced to 0.707 times of maximum Gain.



In General,

Roll-of rate of n^{th} order filter is :

$\boxed{-n\,20 \text{ db/sec.}}$

(7). Design a butterworth 1st order LPF with a cutoff frequency of 1kHz.

Sol:-

$$f_h = \frac{1}{2\pi RC} = 1\times10^3 \text{ Hz}.$$

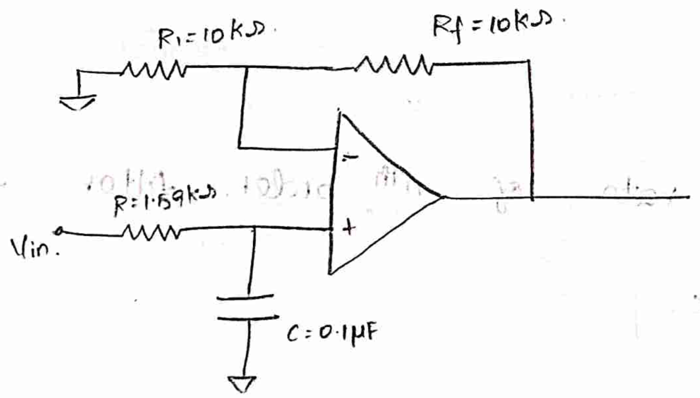
Let us take, $C = 0.1\mu F$.

$$10^3 = \frac{1}{2\times3.14\times R\times0.1\times10^{-6}}.$$

$$R = \frac{1}{6.28\times10^{-7}\times10^3}.$$

$$R = 0.159\times10^4$$

$$\boxed{R = 1.59\text{ k}\Omega.}$$



here, $1+\frac{R_f}{R_1} = A_f = 1.$

assume $R_1 = R_f = 10\text{k}\Omega.$

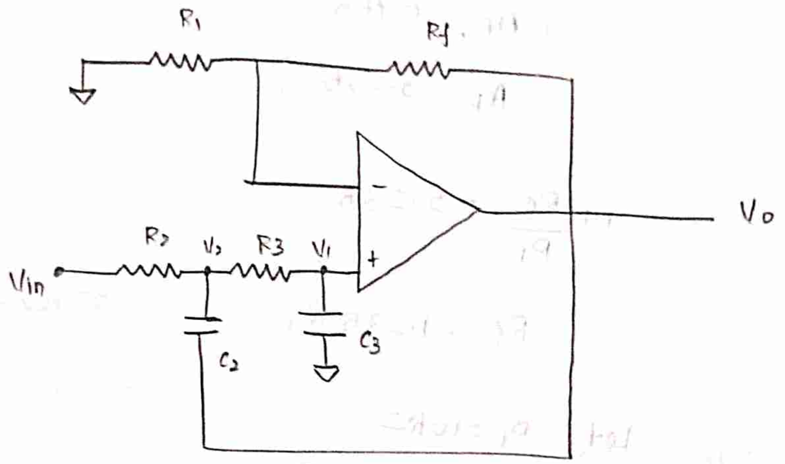
$$\therefore\ 1+\frac{R_f}{R_f} = A_f.$$

$$1+1 = A_f.$$

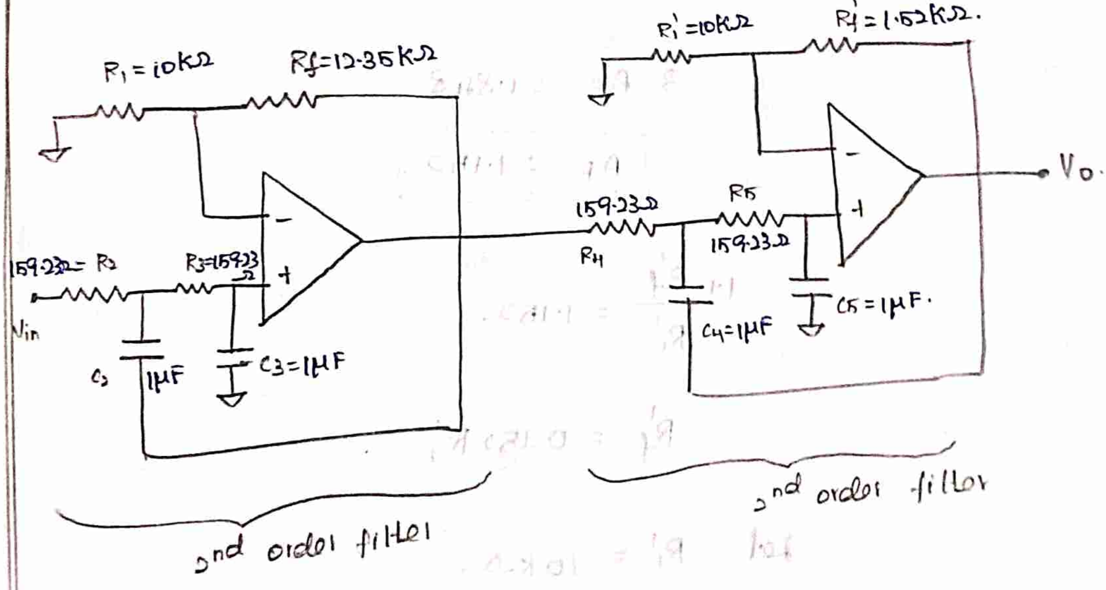
$$\boxed{\text{gain, } A_f = 2.}$$

2^{nd} order Butterworth LPF:

The order of Butterworth LPF which strongly no. of RC section connected in the circuit.



⑧ Design a butterworth 4^{th} order LPF with cutoff frequency of 1KHZ.



Sol: Given, $f_h = 1KHZ$.

W.K.T, $(s^2 + 0.765 s + 1)(s^2 + 1.848 s + 1)$ ↪ ①.

From 2nd order Butterworth LPF,

$$s^2 + \frac{(3-A_F)}{RC} s + \frac{1}{R^2C^2} \rightarrow (2)$$

compare (1) & (2), for 1st 2nd order.

$$3 - A_F = 0.765$$

$$\boxed{A_F = 2.235}$$

$$1 + \frac{R_f}{R_1} = 2.235$$

$$R_f = 1.235 R_1$$

Let $R_1 = 10k\Omega$.

$$\boxed{R_f = 12.35 k\Omega}$$

compare (1) & (3), for 2nd (2nd order)

$$3 - A_F = 1.848$$

$$\boxed{A_F = 1.152}$$

$$1 + \frac{R_f'}{R_1'} = 1.152$$

$$R_f' = 0.152 R_1'$$

Let $R_1' = 10k\Omega$.

$$\boxed{R_f' = 1.52 k\Omega}$$

Let us assume, $R_2 = R_3 = R_4 = R_5 = R$.

$$C_2 = C_3 = C_4 = C_5 = C.$$

$$f_h = \frac{1}{2\pi RC} = 10^3$$

Let, $\boxed{C = 1\mu F.}$

$$R = \frac{1}{6.28 \times 10^3 \times 10^{-6}}$$

$$R = 0.159\ k\Omega.$$

$$\boxed{R = 159.23\ \Omega.}$$

Difference btw 1st & 2nd order LPF:

1). RC sections.

For 1st order – 1 RC section

For 2nd order – 2 RC sections.

2). Roll-of rate.

1st order = -20 db/dec.

2nd order = -40 db/dec.

For n^{th} order LPF:

$$\left|\frac{V_o}{V_{in}}\right| = \frac{1}{\sqrt{1+\left(f/f_h\right)^{2n}}} \quad ; \quad f_h = \frac{1}{2\pi RC}.$$

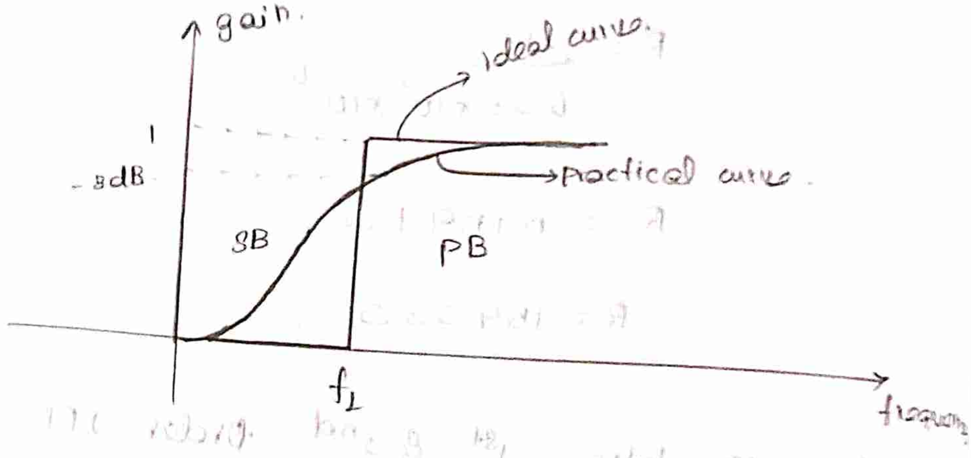
$$\left|\frac{V_o(s)}{V_i(s)}\right| = \left|\frac{V_o(j\omega)}{V_i(j\omega)}\right| = |H(j\omega)|$$

It allows low freq & it eliminates high freq / Noise

[It is used in band-limited signal]

HPF → It allows all the frequency above the cut-off frequency.

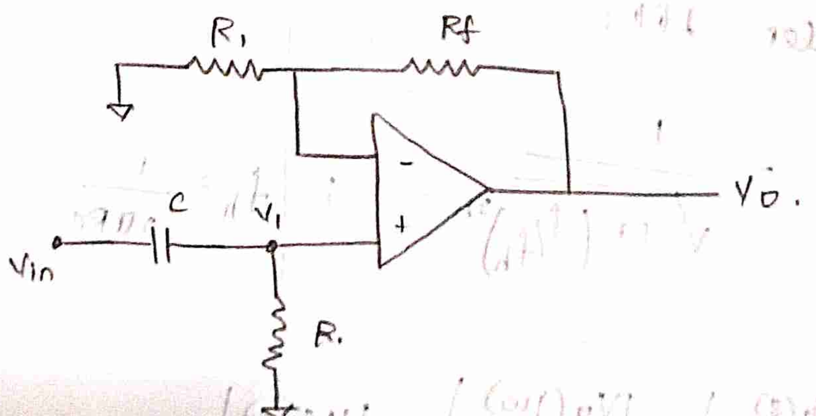
High pass filter:



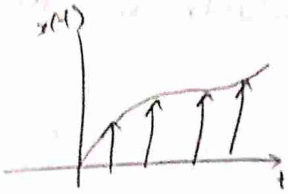
Exchange capacitor & Resistor in LPF we get HPF.

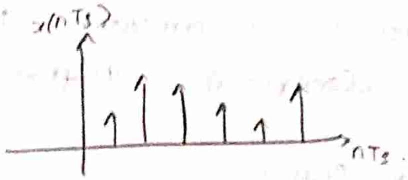
The filter which can allow all signal above the cutoff frequency and attenuate all the frequency signal below the cutoff frequency.

1st order active High pass filter.



$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_1$$





analog

discrete - amplitude is continuous
digital, it has only 0's & 1's

Quantizer - It assigns the binary value to each & every amplitude of signal.

By smoothing the stair signal of i/p (of DAC) signal, we get the approximate transmitted signal.

Sampling & Reconstruction :

⇒ To sample we have 2 condition.

1). Nyquist condition / criterion - $\boxed{f_N = 2f_m = f_c}$.

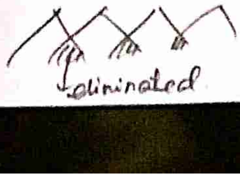
*. It says how to select the sample frequency.

$\boxed{f_s \geq 2f_m}$ [inorder to reconstruct the original transmitter signal.]

2). To sample the analog signal should be band - limited signal.

*. for that we use Anti - alising filter

Aliasing effect - overlapping of Adjacent spectrum because of that improper in nyquist condition, o/p is we get some elimination



Types:

1. weight

2. R-2R

3. 3 mo

DAC :

we

1 o/p.

For DAC

summing

①. weighted